Estimating the sum of a series:

If $f(x)$ is a function such that $f(k)=a_k$, where f is continuous, decreasing, and $\sum_1^\infty a_n$ is convergent. So.

$$\int_{n+1}^{\infty} f(x)\,dx \le R_n \le \int_n^{\infty} f(x)\,dx$$

where $R_n = s - s_n$; $|R_n| = |s - s_n|$ is called the error.

For $\sum_{n=1}^{\infty} \frac{1}{n^3}$ and $n=10$ we have.

$$1.201664 \le s \le 1.202532.$$

If we approximate the sum s by the midpoint of this interval, then the error is at most half the length of the interval. So.

$$\sum_{n=1}^{\infty} \frac{1}{n^3} \approx 1.2021 \quad \text{with error} < 0.005.$$

11.4 Comparison test.

The Direct Comparison test.

Let's consider $\sum a_n$ and $\sum b_n$ are series with positive terms.

i). If $\sum b_n$ is convergent and $a_n \le b_n$ then $\sum a_n$ is convergent.

ii). If $\sum b_n$ is divergent and $b_n \le a_n$ then $\sum a_n$ is divergent.

Example: Determine whether the series $\sum_{n=1}^{\infty} \frac{5}{2n^2+4n+3}$ converges or diverges.

Sol: $b^2-4ac = 16 - 4(2)(3) < 0$. Notice that $\frac{5}{2n^2+4n+3} > 0$ for all n.

let consider $\frac{5}{2n^2+4n+3} \le \frac{5}{2n^2}$.

and $\sum_{n=1}^{\infty} \frac{5}{2n^2} = \frac{5}{2}\sum_{n=1}^{\infty} \frac{1}{n^2} < \infty$ b/c is a p-series with $p=2>1$.

Now, by the comparison thm $\sum_{n=1}^{\infty} \frac{5}{2n^2+4n+3}$ is convergent.

Example. Test the series $\sum_{k=1}^{\infty} \frac{\ln k}{k}$ for convergence or divergence.

Sln: Let consider $\frac{\ln n}{n} \geq \frac{1}{n}$ and since $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent.

So By the comparison thm $\sum_{n=1}^{\infty} \frac{\ln n}{n}$ is divergent.

Limit Comparison test.

Thm: Suppose $\sum a_n$ and $\sum b_n$ are series with positive terms.

If

$$\lim_{n\to\infty} \frac{a_n}{b_n} = c.$$

where c is finite number $c>0$, then either both series converges or both diverge.

Rmk: Make sure that you have in the denominator the general term of the series $\sum b_n$ and we know if $\sum b_n$ is convergence or divergent series.

Example: test the series $\sum_{n=1}^{\infty} \frac{1}{2^n - 1}$ for convergence or divergence.

sln: Let consider $b_n = \frac{1}{2^n} = \left(\frac{1}{2}\right)^n$ and $\sum_{n=1}^{\infty} \frac{1}{2^n} < \infty$ b/c is a geometric series with $a=1$, $r=\frac{1}{2}$ and $|r|<1$.

let $a_n = \frac{1}{2^n - 1}$ and

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{\frac{1}{2^n-1}}{\frac{1}{2^n}} = \lim_{n\to\infty} \frac{2^n}{2^n - 1} = \lim_{n\to\infty} \frac{1}{1 - 1/2^n} = 1$$

so, $1>0$ and By the limit comparison test since $\sum b_n$ is convergent then $\sum_{n=1}^{\infty} \frac{1}{2^n - 1}$ is convergent.

Example: Determinate whether the series $\sum_{n=1}^{\infty} \frac{2n^2+3n}{\sqrt{5+n^5}}$ converges or diverges.

Sol: Let $b_n = \frac{2n^2}{\sqrt{n^5}} = \frac{2}{\sqrt{n}} = \frac{2}{n^{1/2}}$ and $\sum_{n=1}^{\infty} \frac{2}{n^{1/2}}$ is divergent b/c is a p-series with $p = 1/2 < 1$. Now,

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{\frac{2n^2+3n}{\sqrt{5+n^5}}}{\frac{2}{\sqrt{n}}} = \lim_{n\to\infty} \frac{(2n^2+3n)\sqrt{n}}{2\sqrt{5+n^5}}$$

$$= \frac{1}{2} \lim_{n\to\infty} \frac{2n^{5/2} + 3n^{3/2}}{\sqrt{5+n^5}}$$

$$= \frac{1}{2} \lim_{n\to\infty} \frac{2n^{5/2}/n^{5/2} + 3n^{3/2}/n^{5/2}}{\frac{\sqrt{5+n^5}}{n^{5/2}}}$$

$$= \frac{1}{2} \lim_{n\to\infty} \frac{2 + 3/n}{\sqrt{5/n^5 + 1}} = \frac{1}{2}\left(\frac{2+0}{\sqrt{0+1}}\right) = 1 > 0$$

By the limit comparison thm since $\sum b_n$ is divergent so

$$\sum_{n=1}^{\infty} \frac{2n^2+3n}{\sqrt{5+n^5}} \text{ is divergent.}$$

Example: $\sum_{n=1}^{\infty} \frac{n^2-3n}{\sqrt{n^5+2}}$. $a_n = \frac{n^2-3n}{\sqrt{n^5+2}}$ $b_n = \frac{n^2}{\sqrt{n^5}} = \frac{n^2}{n^{5/2}} = \frac{1}{\sqrt{n}}$

and $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} = \sum_{n=1}^{\infty} \frac{1}{n^{1/2}}$ is divergent by p-series with $p = \frac{1}{2} < 1$.

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{\sqrt{n}(n^2-3n)}{\sqrt{n^5+2}} = \lim_{n\to\infty} \frac{n^{5/2} - 3n^{3/2}}{\sqrt{n^5+2}}$$

$$= \lim_{n\to\infty} \frac{(1 - 3/n)}{\sqrt{1 + 2/n^5}} = 1 > 0 \quad \text{By the limit comparison}$$

test $\sum_{n=1}^{\infty} \frac{n^2-3n}{\sqrt{n^5+2}}$ is divergent

Problem: let consider $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^3}$ is alternating series

If $|a_n| \to 0$ as $n \to \infty$ and $|a_n|$ is decreasing. then $\sum_{n=1}^{\infty} a_n$ is convergent

let consider $a_n = \frac{(-1)^n}{n^3}$ then $|a_n| = \frac{1}{n^3} \longrightarrow 0$ as $n \to \infty$.

Notice that $(n+1) \geq n \Rightarrow (n+1)^3 \geq n^3 \Rightarrow \frac{1}{n^3} \geq \frac{1}{(n+1)^3}$

$$|a_n| \geq |a_{n+1}| \text{ for all } n \geq 1$$

So is decreasing. Then $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^3}$ is convergent by alternating test.

We can try approximate using.

$$|\text{error}| = |R_n| = |S - S_n|$$

find the smallest n such that $|\text{error}| < 1/100$

$$|S - S_n| \leq |a_{n+1}| < \frac{1}{100}$$

$$\frac{1}{(n+1)^3} < \frac{1}{100}$$

$$\Rightarrow (n+1)^3 > 100$$

when $n = 4$ we have $(4+1)^3 > 100$ ✓

$n = 3$ we have $(3+1)^3 > 100$ ✗.

So $\boxed{n = 4}$